IMTIAZ AHMED

Email: imti.ahmed19@gmail.com **Location:** Dhaka, Bangladesh
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Electrical and Electronic Engineering graduate with a strong academic focus on renewable energy and advanced photovoltaics. Experienced in working on various projects involving the simulation and optimization of next-generation perovskite solar cells. A dedicated and hardworking individual with a solid foundation in project management and teamwork, eager to leverage my technical skills to contribute to impactful clean energy innovations

EDUCATION

- **Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh**
  Bachelor of Science in Electrical and Electronic Engineering 2020 –2024 (Final Exam: June, 2025): ***CGPA: 3.47 out of 4.0***
- **Dhaka College, Dhaka**
  Higher Secondary Certificate (HSC), 2019: ***Grade: GPA 5.0 out of 5.0***
- **Ideal School and College, Banasree, Rampura, Dhaka-1219**
  Secondary School Certificate (SSC), 2017: ***Grade: GPA 5.0 out of 5.0***

LANGUAGES

- **Bangla**: *Native Language*
- **English**: *Intermediate Listener, Novice Speaker, Advanced Reading and Writing*

PROFESSIONAL EXPERIENCE

- **Research Assistant** [2025-2026]
  Supervisors: Rafiqul Islam, PhD, Professor, EEE, Leading University, Sylhet
  Afshana Begum, Lecturer, EEE, SUST
  Worked on renewable energy mainly focused on designing tandem-solar cell and absorber layer (Perovskite) and DFT analysis.
- **Research Assistant** [2024-2025]
  Supervisor: Dr. Md Rasedujjaman, PhD, Associate Professor, EEE, SUST
  Worked on embedded systems and IoT projects, designing a custom ESP32-S3 datalogger for real-time data acquisition, hardware design, and firmware development.
- **Research Assistant** [2023-2026]
  Supervisor: Arif Ahammad, Assistant Professor, EEE, SUST
  Worked on induction motor fault analysis with embedded machine learning, hybrid solar-wind energy system design, and DFT analysis of transition and non-transition metal-doped perovskite materials.
- **AutoMama** | *Team Leader* [2023–2025]
  Led autonomous vehicle projects, managing teams and driving strategy at AutoMama.
- **RoboSUST** | *Vice President* [2024–2025]
  Managed activities, led workshops, and mentored members for robotics competitions.
- **JLCPCB** | Contributing Writer (Electronics) [2025–2026]
  Wrote technical blog articles on PCB manufacturing, sharing insights as an electronic engineer.

RESEARCH EXPERIENCE

Completed Projects

- **Design and Optimization of Dual Layer Perovskite Solar Cell**
- **DFT analysis for perovskite absorber material.**
- **Induction Motor Fault Analysis**
- **Renewable Energy System for Grid Integration**
- **IoT-Driven Smart Workplace Ecosystem**
- **Custom ESP32-S3 Datalogger Board**

PUBLICATIONS

Journal Papers in Review

- Edge-AI-Driven Real-Time Induction Motor Fault Detection and Predictive Maintenance
- Unveiling Photovoltaic Potential of Lead-free Cs_3SbBiX_9 (X=Cl, Br, I) Perovskite Solar Cells Through DFT Analysis and SCAPS-1D Simulation

Journal Papers

- "MLPNN and Ensemble Learning Algorithm for Transmission Line Fault Classification," International Transactions on Electrical Energy Systems, 2025. doi: 10.1155/etep/6114718.

Conference Papers

- "A Comparative Analysis of Power Consumption and Security Features in LoRa and LoRaWAN Messaging Devices" doi: 10.1109/ICAEEE62219.2024.10561782.
- "IoT-Driven Smart Workplace Ecosystem with RFID Security and Environmental Monitoring Featuring App Integration" doi: 10.1109/PEEIACON63629.2024.10800453.
- "Design and Simulation of a Novel All-Inorganic Dual-Absorber Solar Cell Achieving 31.52% Efficiency" doi: 10.1109/ICECTE69292.2026.11429314

TECHNICAL SKILLS

- **Basic Computer Skill:** *Proficient in Microsoft Office and general computer operations, including file management, internet use, and basic troubleshooting.*
- **Programming:** *Python, Embedded C.*
- **Simulation and Software:** *Quantum Expresso, Material Studio (CASTEP), SCAPS 1D, LTspice, MATLAB, Simulink, AutoCAD, OriginPro, Proteus, MicroWind, EMU8086, Arduino IDE, Overleaf, Arduino, Platform IO.*
- **Machine Learning:** *TensorFlow, Edge Impulse, Sci-kit Learn, TinyML.*
- **PCB Design:** *Proficient in PCB design for embedded and IoT applications using EasyEDA.*

Soft SKILLS

Hardworking and dedicated, Attentive listener, Quick learner and adaptable, Strong-communication skills, Teamwork and collaboration, Problem-solving mindset

COMPETITIONS AND ACHIEVEMENTS

- **Champion, JRC Board Hackathon 2023**
- **Runner-Up, AGP 2024 (Line Following Robot Contest)**
- **Champion, AGP 2022**
- **University Innovation Hub Program (5th Place)**

CERTIFICATION AND TRAINING

- **TICI Certification**: 21 days industrial training on Electrical Power Systems, Power Control, Instrumentation, and PLC. Achieved A+ Grade**.**
- **Conference Presentation Certificate**
    - **ICAEEE**: Presented research on power consumption and security in LoRa and LoRaWAN at ICAEEE 2024.

REFERENCES

1. **Dr. Md Rasedujjaman, PhD**
   Associate Professor
   Department of Electrical and Electronic Engineering
   Shahjalal University of Science and Technology, Sylhet, Bangladesh
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2. **Arif Ahammad**
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